



## Using Altman's Z-score

### Case

This case was written by Dr. Jasbir Singh Matharu, Associate Professor (Finance) at Institute of Management Technology, Nagpur, India. It is intended to be used as the basis for class discussion rather than to illustrate either effective or ineffective handling of a management situation. The case note was compiled from published sources.

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It was the evening of 25 March 2020 in Mumbai. Raj Mehta, who had recently completed his MBA degree, came across a news item from 2019 that said Jet Airways had suspended its operations because it could not pay its massive debt. Raj had recently started investing in stock markets. With this news, he was perplexed about the future of the companies he wanted to invest in. He wanted to find out if there was a way to predict bankruptcy.

He contacted his friend Abhi, who has been in the field of investments for the last four years. They met in a café near Raj's house. Abhi told Raj, "I have heard of a model called Altman's Z-score, but I have not used it. I will try to send you some material."

He sent the following material to Raj:

*Professor Edward Altman had created a model in 1968 using discriminant analysis for publicly traded large manufacturing firms. This model attempts to predict bankruptcy in the next two years. He later modified the model to suit private companies and service firms.*

*The equation below calculates the Z-score:*

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

*The variables X1 through X5 are five ratios that can be computed using the financial statements of a firm.*

$$X_1 = \frac{\text{Net Working Capital}}{\text{Total Assets}}$$

$$X_2 = \frac{\text{Retained Earnings}}{\text{Total Assets}}$$

$$X_3 = \frac{\text{Earnings before Interest and Taxes}}{\text{Total Assets}}$$

$$X_4 = \frac{\text{Market Value of Equity}}{\text{Book Value of Liabilities}}$$

$$X_5 = \frac{\text{Sales}}{\text{Total Assets}}$$

*If the Z-score is above 2.675, the firm is safe. If the score is below 1.81, there is a very high possibility of the firm experiencing bankruptcy. A score between 1.81 and 2.675 may indicate a possibility of bankruptcy.*

Along with this material, Abhi sent the Balance Sheet and Profit and Loss Statement of the recently published annual report of ACC Limited. (Exhibit 1 and 2). He also mentioned that the market capitalization of ACC Limited as of 31 December 2019 was INR 271,474.66 million.

Raj went through the material and the financial statements but could not calculate the Z-score. He thought of calling his cousin, Ravi Grover, who worked in an investment banking firm.

## Exhibit 1

**BALANCE SHEET of ACC Limited**  
**as at 31 December 2019**

| <b>Particulars</b>                                                                | <b>INR Million</b> |
|-----------------------------------------------------------------------------------|--------------------|
| <b>ASSETS</b>                                                                     |                    |
| <b>Non-Current Assets</b>                                                         |                    |
| Property Plant and Equipment                                                      | 69572.80           |
| Capital Work in Progress                                                          | 4353.40            |
| Other Intangible Assets                                                           | 340.90             |
| Investments in subsidiaries, associates and joint ventures                        | 2264.50            |
| Financial Assets                                                                  |                    |
| <i>Investments</i>                                                                | 37.00              |
| <i>Loans</i>                                                                      | 1359.20            |
| <i>Other Financial Assets</i>                                                     | 4682.30            |
| Non-Current Tax Assets (Net)                                                      | 8570.10            |
| Other non-current assets                                                          | 5407.80            |
| <b>Total Non-current assets</b>                                                   | <b>96588.00</b>    |
| <b>Current Assets</b>                                                             |                    |
| Inventories                                                                       | 11409.50           |
| Financial Assets                                                                  |                    |
| <i>Trade Receivables</i>                                                          | 6284.30            |
| <i>Cash and cash equivalents</i>                                                  | 43831.80           |
| <i>Bank Balances</i>                                                              | 1549.20            |
| <i>Loans</i>                                                                      | 314.30             |
| <i>Other Financial Assets</i>                                                     | 2705.10            |
| Other current assets                                                              | 8034.10            |
| Non-current assets classified as held for sale                                    | 104.70             |
| <b>Total Current assets</b>                                                       | <b>74233.00</b>    |
| <b>TOTAL ASSETS</b>                                                               | <b>170821.00</b>   |
| <b>EQUITY AND LIABILITIES</b>                                                     |                    |
| Equity                                                                            |                    |
| Equity Share Capital                                                              | 1879.90            |
| Other Equity                                                                      | 113332.90          |
| <b>Total Equity</b>                                                               | <b>115212.80</b>   |
| Liabilities                                                                       |                    |
| Non-current liabilities                                                           |                    |
| Provisions                                                                        | 2341.30            |
| Deferred tax liabilities (net)                                                    | 6422.10            |
| <b>Total Non-current liabilities</b>                                              | <b>8763.40</b>     |
| Current Liabilities                                                               |                    |
| Financial Liabilities                                                             |                    |
| <i>Trade Payables</i>                                                             |                    |
| <i>Total outstanding dues of micro and small enterprises</i>                      | 112.70             |
| <i>Total outstanding dues of creditors other than micro and small enterprises</i> | 14597.00           |
| <i>Other financial liabilities</i>                                                | 9339.60            |
| Other current liabilities                                                         | 19138.00           |
| Provisions                                                                        | 233.90             |
| Current Tax liabilities (net)                                                     | 3423.60            |
| <b>Total Current liabilities</b>                                                  | <b>46844.80</b>    |
| <b>Total Liabilities</b>                                                          | <b>55608.20</b>    |
| <b>TOTAL - EQUITY AND LIABILITIES</b>                                             | <b>170821.00</b>   |

Source: <https://www.acclimited.com/newsite/annualreport2019/Chapter/Standalone-Financial-Statements.pdf>

## Exhibit 2

**STATEMENT OF PROFIT AND LOSS of ACC Limited**

for the year ended December 31, 2019

| <b>Particulars</b>                                                             | <b>INR Million</b> | <b>INR Million</b> |
|--------------------------------------------------------------------------------|--------------------|--------------------|
| <b>INCOME</b>                                                                  |                    |                    |
| 1 Revenue from operations                                                      | 156566.50          |                    |
| 2 Other income                                                                 | 3112.10            |                    |
| <b>3 Total Income (1+2)</b>                                                    |                    | 159678.60          |
| <b>4 EXPENSES</b>                                                              |                    |                    |
| Cost of Material Consumed                                                      | 22581.00           |                    |
| Purchase of stock-in-trade                                                     | 3616.90            |                    |
| Changes in inventories of finished goods, work-in-progress, and stock-in-trade | 1008.10            |                    |
| Employee benefits expense                                                      | 8639.70            |                    |
| Power and fuel                                                                 | 31313.40           |                    |
| Freight and Forwarding expense                                                 | 40500.60           |                    |
| Finance costs                                                                  | 862.20             |                    |
| Depreciation and amortisation expense                                          | 6029.70            |                    |
| Other expenses                                                                 | 24835.50           |                    |
|                                                                                | 139387.10          |                    |
| Captive Consumption of Cement                                                  | -23.20             |                    |
| <b>Total Expenses</b>                                                          |                    | 139363.90          |
| <b>5 Profit before tax (3-4)</b>                                               |                    | 20314.70           |
| <b>6 Tax expense</b>                                                           |                    |                    |
| Current tax                                                                    | 6898.10            |                    |
| Deferred tax (credit) / charge                                                 | -172.50            |                    |
|                                                                                |                    | 6725.60            |
| <b>7 Profit for the year (5-6)</b>                                             |                    | <b>13589.10</b>    |
| 8 Other Comprehensive Income (OCI)                                             |                    |                    |
| Items that will not be reclassified to profit and loss:                        |                    |                    |
| Re-measurement gain / (loss) on defined benefit plans                          |                    | -752.80            |
| Income tax relating to items that will not be reclassified to profit and loss  |                    | 263.00             |
| <b>Other Comprehensive Income for the year, net of tax</b>                     |                    | <b>-489.80</b>     |
| <b>9 Total Comprehensive Income for the year (7+8)</b>                         |                    | <b>13099.30</b>    |
| <b>10 Earnings per equity share of INR 10 each</b>                             |                    |                    |
| Basic                                                                          |                    | 72.36              |
| Diluted                                                                        |                    | 72.19              |

Source: <https://www.acclimited.com/newsite/annualreport2019/Chapter/Stand-alone-Financial-Statements.pdf>

**Reference**

E. Altman, "Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy," Journal of Finance, September 1968, pp. 589–609.